What eQTL Data Is

And how a meiotic mosaic fits an ontology built for gene edits

Michael Volk

September 12, 2026

Abstract

Expression quantitative trait locus studies cross two strains, collect recombinant progeny, and measure a transcriptome in every one. Four of them rank among the strongest candidates for ingestion, and three are the only datasets in that list that are high-dimensional in both perturbation and readout. They also break an assumption the current schema rests on: the genotype is never observed directly. It is inferred from sparse markers as a mosaic of two parental genomes. This sets out what the records contain, establishes that the strain genotype is nevertheless recoverable to roughly one percent while the causal variant is not recoverable at all, and states what the schema needs before a loader is written.

Section status: ✓ ready ■ author-reviewed, keep stable ✗ not done, agents may edit freely

Provenance: ▲ not in the library mirror; verify at the linked source ● from a review's summary table, not the primary paper

Contents

1 The experiment and its three tables ✗

Two strains that differ genetically are crossed, classically BY, a laboratory strain near-isogenic to S288C, against RM, a vineyard isolate [1]. The F1 hybrid is sporulated and n haploid progeny are collected. Each progeny strain, a **segregant**, is a **meiotic mosaic**: at every locus it carries either the BY or the RM allele, in linked blocks whose boundaries are set by crossover positions. Those boundaries are not known ahead of time: every segregant draws its own crossover positions in meiosis, so they must be inferred per strain from the genotype data, and locating them is where the genotype error concentrates (Sec. 2). Every segregant's transcriptome is measured, and the question asked per gene is which loci in the mosaic predict its expression.

The perturbation is therefore not applied to a strain. It is inherited, and it is present everywhere in the genome at once.

1.1 Two measurements and one inference ✗

Genotype matrix. X, segregants by marker positions. Entries encode *parental origin* rather than base identity, so the natural form looks binary. Bloom 2013 used 11,623 markers for 1,008 segregants [2], and that older panel is the deliberate reference point rather than a stale one: the single-cell study of Sec. 2 drew its strains from this same cross, pooling "393 previously genotyped haploid segregants" [3]. BY and RM segregate at roughly 4.5×10^4 sites: sites where the two parents carry different alleles, which therefore separate into different progeny at meiosis (Mendelian segregation) and vary across the panel. Nothing is swapped or edited at these sites; segregation is inheritance, not an applied change. The marker set is a thinned representation of that variant set, and long runs of constant value along the position axis are what recombination blocks look like. Sec. 2 corrects the binary reading: the entries the mapping models actually consume are posterior probabilities, not calls.

Expression matrix. $Y \in \mathbb{R}^{n \times p}$, segregants by genes, with $p \approx 5{,}000$ to $6{,}000$. This is the phenotype, and it is a vector per strain rather than a scalar.

QTL table. The derived output. For gene g and marker j the standard model is

$$y_{ig} \;=\; \mu_g \;+\; \beta_{gj}\, X_{ij} \;+\; \varepsilon_{ig}, \qquad \varepsilon_{ig} \sim \mathcal{N}(0, \sigma_g^2)$$

fit marker by marker, with significance from a LOD score, the $\log_{10}$ odds of the data under linkage between marker and expression level against the null of no linkage, and a permutation threshold. A row of the table is a gene, a marker interval, an effect size, a LOD and a variance explained. Panel d of Fig. 1 is this table drawn as a matrix.

The first two are measurements. The third is an inference over them, and it changes if the mapping method, the threshold or the marker density changes while the measurement does not. Sec. 3 returns to what that means for storage.

1.2 Structure worth knowing before modeling it ✗

Cis against trans. A **cis** eQTL sits at or near the gene's own locus and is usually a promoter variant changing that gene's own transcription. A **trans** eQTL sits elsewhere, typically in a regulator. Cis effects are strong, sparse and roughly one per affected gene; trans effects are weak and numerous.

Hotspots. Trans effects are not spread evenly across loci. A small number of loci each affect hundreds to thousands of transcripts, one regulatory variant propagating outward. The columns of Y are therefore strongly dependent, a few latent factors explain much of the variance, and treating genes as independent is wrong. This hotspot structure is also the leverage genome-scale engineering aims at: a single regulatory locus that moves hundreds of transcripts.

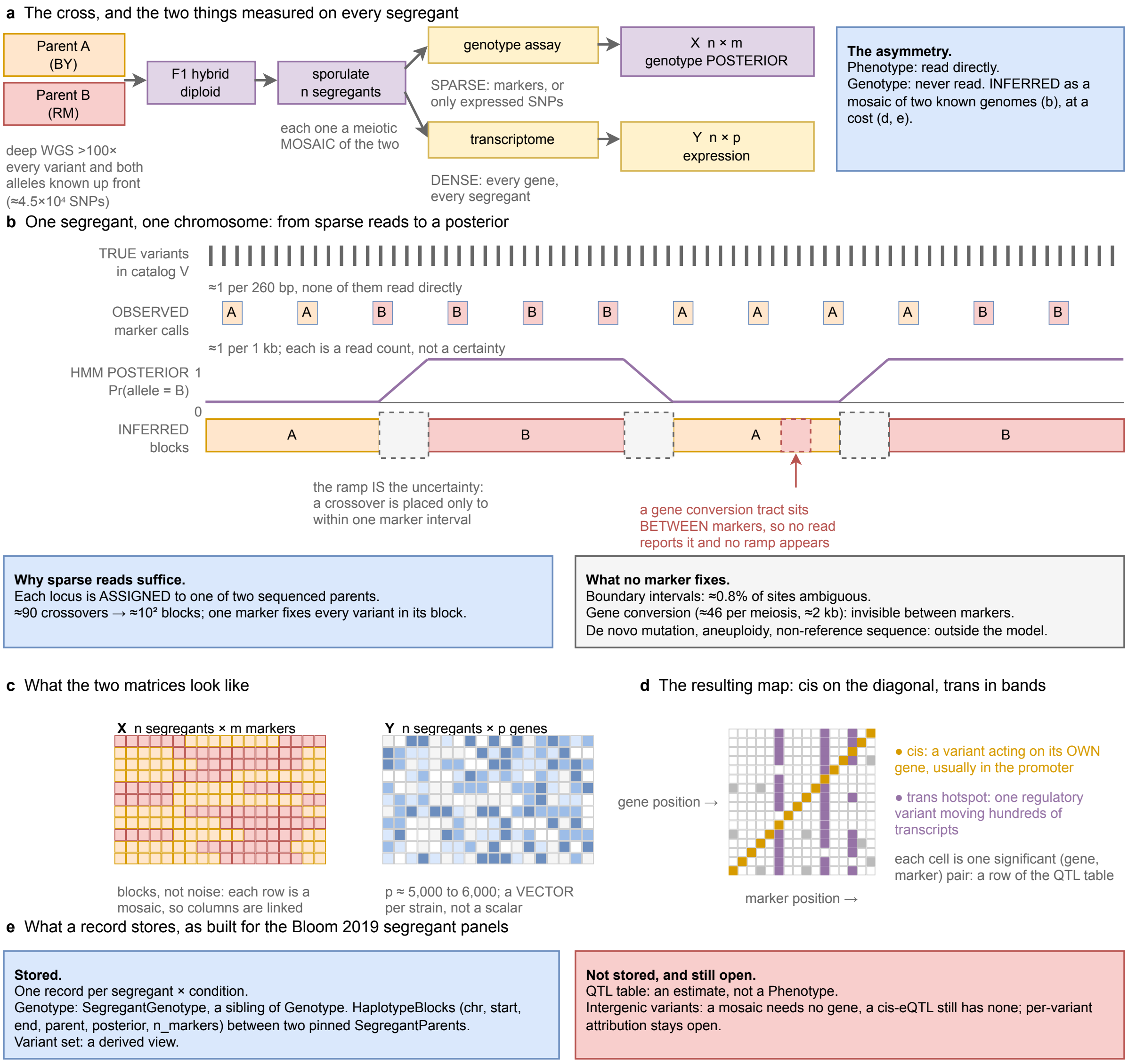

Figure 1: An eQTL study, where inference enters it, and what it costs. **a**, The cross. Both parents are deep-sequenced, so every variant and both its alleles are known before any segregant is examined. The segregants are then assayed twice, sparsely for genotype and densely for expression, and that asymmetry drives everything else. **b**, One segregant along one chromosome, at four levels. True variant sites are dense and none is read directly; marker calls are sparse and are read counts rather than certainties; the HMM output is a posterior whose ramps are the uncertainty, since a crossover is placed only to within one marker interval; the inferred blocks follow. A non-crossover gene conversion tract sitting between markers produces no ramp at all, because no read reports it. Sparse reads suffice because the segregant is never sequenced de novo: at each locus it is assigned to one of two already-sequenced genomes, meiosis makes about 90 crossovers, so a genome is about 10^2 blocks and one marker fixes every cataloged variant in its block; the task is locating about 10^2 boundaries, not determining about 10^4 values. What no marker density fixes: boundary intervals leave about 0.8% of cataloged sites ambiguous; non-crossover gene conversion (about 46 tracts per meiosis, about 2 kb each) affects more bases than that and is invisible to sparse markers; de novo mutation, aneuploidy (a haploid can carry a disomy) and non-reference sequence sit outside the mosaic model entirely. **c**, The two matrices. X shows haplotype blocks rather than noise, so its columns are linked; Y is a vector per strain rather than a scalar. **d**, The map that results, and it is the QTL table of Sec. 1.1 drawn as a matrix: each cell is one significant (gene, marker) row of that table, plotted gene position against marker position. Cis effects fall on the diagonal, each variant acting on its own gene and usually through its promoter; trans hotspots appear as vertical bands where one regulatory variant moves hundreds of transcripts. **e**, What a record stores, as built for the Bloom 2019 segregant panels (Sec. 3.6). Stored: one record per segregant per condition, the pair (genotype, environment) to phenotype; the genotype is a `SegregantGenotype`, a sibling of `Genotype` rather than a perturbation leaf, holding haplotype blocks of (chromosome, start, end, parent, posterior, marker count) between two `SegregantParents` pinned to sha256-anchored assemblies, so the variant set is a derived view and the posterior is stored as a posterior. On the built store the blocks per segregant run from a median of 83 (cross A) to 143 (medians by cross), with tails to 1,847 from hard calls at posterior 1.0. Not stored: the QTL table, an estimate conditional on method, threshold and marker density with no source value to check, so it is not a `Phenotype`. Still open: a mosaic needs no `systematic_gene_name` because it is not a `GenePerturbation`, but a cis-eQTL is usually an intergenic promoter variant with no gene to attach to, so per-variant attribution and the reporter-assay class remain one decision. Panels **b** and **d** are schematic rather than plotted from data. Composed from the sources in Sec. 1 rather than redrawn from a published figure; source `notes/assets/drawio/eqtl-experiment-and-genotype-inference.drawio`, emitted by `notes/assets/drawio/eqtl-experiment-and-genotype-inference.gen.py`.

1.3 The four candidates, and why they rank where they do ✗

Table 1: The eQTL rows in the candidate list. *Both axes* marks a dataset that is high-dimensional in perturbation and in readout at once, which in that list is true of three datasets in total.

Dataset	Segregants	Readout	Both axes
Albert 2018 [4]	1,012	bulk RNA-seq, 1 condition	yes
Boocock 2025 [3]	393	one-pot single cell, 2 carbon sources	yes
N'Guessan 2025 [5]	∼4,500 profiled	single cell	yes
Smith 2008 [6]	109	bulk, glucose and ethanol	no

Boocock 2025 draws 27,744 single cells from those 393 segregants, a median of 17 cells per segregant. The extra axis is cells *within* a genotype, which is what allows eQTLs to be mapped for expression noise and for cell-cycle occupancy rather than only for the mean. Smith 2008 is small, and it is the design that demonstrates an eQTL present in one carbon source and absent in the other.

2 The genotype is inferred, not observed ✗

2.1 What the methods actually say ✗

Verbatim from the mirrored Boocock 2025 paper:

> "We used a hidden Markov model (HMM) to reconstruct the patterns of inheritance of parental alleles in each cell based on the observed genotypes at expressed SNPs."

> "We defined prior genotype probabilities as 0.5 for each parental variant. Emission probabilities were calculated as previously described for low coverage sequencing data."

> "Genotype probabilities were standardized, and markers in very high LD ($r > 0.999$) were pruned. This LD pruning is approximately equivalent to using markers spaced 4 centimorgans (cm) apart."

Two terms in that last quote: **LD**, linkage disequilibrium, is the statistical correlation between alleles at different sites, near-perfect for neighboring sites here because they are inherited together in blocks; a **centimorgan** (cM) is a unit of genetic map distance, two loci sitting 1 cM apart when 1% of meioses place a crossover between them.

So the genotype matrix is not binary. It is

$$X \in [0,1]^{n\times m}, \qquad X_{ij} \;=\; \Pr\big(\text{segregant } i \text{ carries the RM allele at } j \,\big|\, \text{reads}\big)$$

and the eQTL model is fit on the standardized posterior directly. Binarizing at 0.5 is done only to compare cell identities.

2.2 Three levels of partial observation ✗

Table 2: What is actually observed, by study. The single-cell design is the most severely partial: genotype coverage is conditioned on the transcript being expressed, so the genotype observation and the phenotype measurement come from the same reads.

Study	Observed	Genotype is
Bloom 2013 [2]	low-coverage WGS of each segregant	calls at ∼11,623 markers
Nguyen Ba 2022 [7]	low-coverage WGS plus barcode	calls at markers, $n \sim 10^5$
Boocock 2025 [3]	only SNPs that happen to be expressed	an HMM posterior per marker

The observation channel differs per study, so the probabilistic genotype has to be extracted differently per study: there is no shared genotype-recovery path for a loader to reuse, only a shared representation for its output.

A gene that is switched off in a cell contributes no genotype information about its own locus. That coupling has no analog in any other dataset in the collection, it is inherent to the one-pot design, and the record can only carry it, not correct it.

2.3 Why sparse observation is nevertheless enough ✗

The segregant is not being sequenced de novo. It is being *assigned* to one of two known genomes at each locus, which is a far easier problem:

1. The parents are deep-sequenced, "deep ($> 100\times$) paired-end sequenced data for our parental strains," so the complete variant catalog and both alleles at every site are known before any segregant is looked at.
2. Meiosis produces few boundaries. In *S. cerevisiae* roughly 90 crossovers per meiosis [8], so across 16 chromosomes a segregant genome is on the order of 10^2 contiguous blocks.
3. Within a block, parental origin is constant, so one marker fixes every cataloged variant in its block.

Reconstruction reduces to locating $\sim 10^2$ boundaries rather than determining $\sim 10^4$ independent values.

Error budget, derived from published parameters rather than measured. BY and RM segregate at $\sim 4.5\times 10^4$ SNPs over 12 Mb, about 1 per 260 bp. Bloom 2013's 11,623 markers give ~ 1 kb spacing, so a crossover is localized to about 4 variants:

$$\underbrace{90}_{\text{crossovers}} \times \underbrace{4}_{\text{variants per interval}} \approx 360 \text{ of } 4.5 \times 10^4 \approx 0.8\% \text{ ambiguous}$$

leaving roughly 99% of cataloged variant sites assignable. ▲ [external: arithmetic from published crossover and variant-density figures, not a measured concordance]

The larger error source is not crossovers. Yeast meiosis also produces ~ 46 non-crossover **gene conversion** events per meiosis with tracts of order 2 kb [8]. A gene conversion is a byproduct of meiotic recombination repair: a double-strand break is repaired by copying sequence from the homologous chromosome, so a short tract of one parent's sequence is overwritten with the other parent's, without exchanging the flanking arms. The result is a ~ 2 kb patch of the "wrong" parent inside an otherwise intact block. Those patches are short, they do not shift the surrounding haplotype, and sparse markers miss them entirely, so they affect more base pairs than boundary uncertainty and are harder to detect.

2.4 What reconstruction cannot give you at any marker density ✗

- **De novo mutations.** A segregant is a mosaic plus whatever arose since the cross. Few per strain, invisible to a parental-assignment method, so the genome is never exactly a mosaic.
- **Aneuploidy and copy number.** Haploidy does not preclude these: a haploid segregant can carry a second copy of a whole chromosome (a disomy) or a segmental amplification, and yeast segregants do acquire them. Detecting that needs read coverage, not marker identity. The mirrored Boocock 2025 text reports no aneuploidy or copy-number screen, so for that design it is an unmodeled error term rather than a measured one.
- **Non-reference sequence.** A variant call file is not an assembly. Reconstructing total genomic content needs the parents' assemblies, including sequence absent from S288C. The ingredients exist: the deep paired-end parental data supports de novo assembly, with a known residual error rate, so holding pinned parent assemblies is an attemptable requirement rather than a new sequencing campaign.

2.5 Per cell the answer flips ✗

Boocock reports *median genotype agreement of 92.5%* between the single-cell HMM genotypes and whole-genome sequencing of the same strains, improving with UMI count. About 1 marker in 13 wrong is entirely acceptable for an association fit across 27,744 cells, where the posterior enters as a standardized covariate and the error averages out. It is nowhere near sufficient to write down a genome. What rescues the strain-level genotype is pooling: a median of 17 cells per segregant, against 393 segregants that had already been whole-genome sequenced in Bloom 2013 and against which the cells were matched. The record never needs the per-cell genotype to be a genome; what it stores is the strain-level mosaic, and that is the same compositional form the schema already uses everywhere else, a pinned reference plus recorded differences, so the insufficiency of a single cell's genotype does not threaten the ingestion pattern.

Do not conflate reconstruction resolution with mapping resolution. The HMM used all expressed SNPs. The 4 cM LD-pruned marker set was used for the association, because markers above $r > 0.999$ carry no independent information. Coarse mapping resolution says nothing about how well the genome can be reconstructed.

2.6 Reconstruction and attribution come apart ✗

Two questions about the same data sound alike and have opposite answers.

Reconstruction succeeds. *Which parent's allele does segregant i carry at site j?* This is a two-way choice between two fully sequenced genomes, decided jointly for a whole block by every marker in it, and the arithmetic of Sec. 2.3 puts it at ∼99% of sites. More segregants, denser markers or deeper coverage all push it further toward certainty.

Attribution is blocked. *Which variant causes the expression difference?* Within a haplotype block every variant is inherited with every other, so their columns in X are identical and no fit can tell them apart. The effect is identifiable only up to the block. Collecting more segregants does not help, because the collinearity comes from the cross itself: the same meioses that made the panel made the blocks. Boocock's pruning of markers above $r > 0.999$ concedes exactly this point; markers that co-segregate carry no separate information.

What each answer decides. Reconstruction is what the sequence gate on the candidate list asks about: can the total genomic content of the strain be estimated? It can, so **segregant panels pass**, conditional on holding the parent assemblies. Attribution is what modeling wants and does not get: a cross localizes an effect to an interval and nothing more. Getting from an interval to a base takes an assay that tests alleles one at a time, such as a massively parallel reporter assay, which is why those datasets belong in the same ingestion wave rather than a later one.

3 How it fits the ontology ✗

3.1 What already exists ✗

The primary record is a genotype and an environment mapped to a phenotype, which is exactly `Experiment`. Nothing conceptual is missing, and four pieces are already built.

- `RNASeqExpressionPhenotype` is the bulk readout, and it was written for this situation. Its docstring describes a survey "where each isolate's transcriptome is measured on its OWN genome," with "no common reference to ratio against," storing absolute TPM. Albert 2018 and Smith 2008 fit unchanged.
- `PseudobulkExpressionPhenotype` is the single-cell readout, built for Nadal-Ribelles 2025. It carries `dispersion` and `n_cells` per genotype, which is the right shape for Boocock's noise eQTLs: heterogeneity survives as a per-genotype scalar rather than being averaged away.
- `SequenceVariantPerturbation` states the intent outright. Together with `CopyNumberVariantPerturbation` it "lets a NATURAL ISOLATE be modeled as a perturbation set off S288C (∼4,500 sequence variants per isolate)."
- The off-graph pointer pattern on that class, `sequence_source`, `sequence_uri` and `sequence_sha256`, means variant sequences are never inlined.

3.2 Where the obvious mapping fails ✗

Reusing `SequenceVariantPerturbation` for a segregant is the obvious move and it is wrong. **A natural isolate and a segregant are not the same kind of object.** Caudal's isolates are each individually sequenced, so the variant set is observed and every allele dereferences to a real sequence with a hash. A segregant is never sequenced to that standard; the *parents* are, and the segregant is imputed as a mosaic of them. That class promises a dereferenceable, hash-verified sequence, and for a segregant no such artifact exists. Using it would encode an inference as an observation, which is the precise failure the verification levels exist to prevent.

3.3 Four gaps, in order of how much they matter ✗

Intergenic variants have no representation. `GenePerturbation` requires `systematic_gene_name` and validates it against

```
^(Y[A-P][LR]\d{3}[WC](-[A-Z])?|Q\d{4}|YNC[A-Q]\d{4}[WC])$
```

so every perturbation must attach to a real gene. A promoter SNP 300 bp upstream of an ORF has no gene to attach to, and assigning it to the downstream gene is an interpretation, wrong whenever the variant acts on a divergent neighbor. This is the sharpest gap and it is not incidental: the promoter is where the strongest and most interpretable cis signal lives. It is also the same gap that blocks every massively parallel reporter assay, which perturbs regulatory sequence exclusively. A gene-anchored patch does exist: the genome machinery already carries a flanking-window convention for gene sequence (1,000 bp upstream, 300 bp downstream), which covers most promoter variants, and the anchor need not even be a natural gene, since a synthetic locus name tied to a sequence hash could stand in for an edited or renamed region. Either way the window assignment is an interpretation and would have to be recorded as one.

Cardinality. An SGA record carries at most three perturbations. A segregant carries $\sim 10^4$. For $n \sim 10^3$ segregants that is $\sim 10^7$ perturbation objects for one dataset, and `Genotype` sorts and set-compares its perturbation list on every equality check while the L1 key hashes the genotype signature.

Reference asymmetry. Reference here means the genome that variants are expressed against, not the baseline a phenotype is ratioed to; a record has both, and this gap concerns only the first. A segregant's natural genomic reference is one of its two parents, but the convention expresses every genotype off S288C. BY is near-isogenic to S288C, so BY-derived blocks are nearly empty perturbation sets while RM-derived blocks carry the full variant load. Keeping S288C is still right, since it is what makes these records comparable to everything else, but the lopsidedness reflects the reference choice rather than the biology and should be documented rather than discovered. The phenotype side is untouched by this: expression is stored absolute, with no wild-type ratio in these designs at all.

Linkage is destroyed by a set. Variants arrive in blocks and co-segregate. `Genotype` does hold its perturbations in a stable order, but that order is a canonical sort by systematic gene name and perturbation type, imposed so that equality and hashing are well defined; it is set semantics with deterministic iteration, and lexical gene-name order is not chromosomal order. Nothing in the representation says two variants sit on the same block and travel together, which is exactly the structure that makes mapping possible and that limits attribution.

3.4 The representation that follows ✗

Store what was actually determined, a **haplotype mosaic** rather than a variant set:

$$G_i = \{(\text{chr, start, end, parent}, p)\}, \qquad |G_i| \sim 10^1\text{–}10^2$$

against two hash-pinned parent assemblies, with a per-interval posterior p and the marker spacing recorded as the resolution. This is better on four counts at once.

1. **Honest.** It stores the inference as an inference, with its uncertainty, rather than promising sequence that was never read.
2. **Compact.** Tens of intervals per strain instead of $\sim 10^4$ variants, which dissolves the cardinality problem rather than solving it.
3. **Lossless.** The full variant set is recoverable exactly from the two parent assemblies plus the intervals, so it becomes a derived view and nothing is given up.
4. **Linkage-preserving.** The interval is the unit, so co-segregation is explicit.

The cost is a genuinely new perturbation type, keyed on a genomic interval and a parent rather than on a gene, carrying a probability. That cost is accepted: the type is judged necessary, not merely convenient. It is also the same non-gene-keyed requirement the promoter-variant gap already imposes, so the two should be resolved together.

3.5 What to do ✗

1. **Ingest the two matrices, not the QTL table.** One `RNASeqExpressionExperiment` per segregant per condition, or `PseudobulkExpressionExperiment` for the single-cell studies. The QTL table is an estimate conditional on method and threshold, it has no source value to check against at any verification level, and storing it as a `Phenotype` would be ingesting somebody's analysis as data.
2. **Decide the intergenic question deliberately**, because it gates both this class and the regulatory-DNA class. With the mosaic type accepted, an interval-keyed perturbation leaf exists regardless, so the live options are: attach promoter variants to that same interval-keyed leaf; or keep gene-keying and assign each variant to a gene via the existing flanking-window convention (1,000 bp upstream, 300 bp downstream), with the attribution recorded as an interpretation, e.g. a `ProvenanceGap`; or restrict to coding variants. The third is the honest short-term option and the worst long-term one.

3. **Treat the parent assemblies as an ingestion dependency.** A segregant record means nothing without them, so they are part of the artifact rather than context, and each needs a hash-pinned assembly rather than a variant call file.

4. **Measure the cardinality cost** on a single Albert 2018 segregant before building anything.

3.6 What was built ✗

The mosaic was implemented on 2026-09-12 for the Bloom 2019 segregant panels (eLife 8:e49212, doi 10.7554/eLife.49212; 16 crosses, 13,950 haploid segregants, 38 growth conditions), in `torchcell/datasets/scerevisiae/bloom2019.py`, and it departs from Sec. 3.4 in one respect. The proposal called the cost "a genuinely new perturbation type". A perturbation leaf cannot carry a mosaic: `Genotype` sorts its perturbations on `systematic_gene_name` and every gene-keyed consumer walks that list, so a leaf with no gene either fails the regex or has to carry a name it does not have, and a `Genotype` with an empty list reads as wild-type S288C to every consumer that reads only the list. The mosaic is therefore a *sibling* of `Genotype`: `SegregantGenotype` holds `HaplotypeBlocks` of (chromosome, start, end, parent, posterior, n_markers) between two `SegregantParents` pinned to sha256-anchored assemblies, and `SegregantGrowthExperiment` declares that class as its genotype. The base `Experiment` rejects a segregant dict, so the union resolves the class unambiguously and a gene-keyed consumer fails loudly rather than reading a segregant as unperturbed. The new classes enter only the module-level unions and maps, so no served dataset's schema contract moved and the admission gate reported the increment admissible.

Three of the four properties in Sec. 3.4 held as stated on the built store. Honest: the release is the R/qtl hard call, so every block carries posterior 1.0 and the call method is quoted from the authors' code. Lossless: every one of the 13,950 block sets re-expands to its released marker row, checked at L2. Linkage-preserving by construction. Compact needs a correction. Cross A, the BY × RM cross the estimate of Sec. 3.4 was made for, has a median of 83 blocks per segregant, inside the 10^1–10^2 range; the other 15 crosses run from 90 to 143 at the median, and the means are pulled to 100–240 by tails of segregants with hundreds to 1,847 blocks. A segregant is still a few kilobytes rather than $\sim 10^4$ variants, so the cardinality argument stands, but the block count is not a property of meiosis alone. It is a hypothesis, untested here, that the tail is short runs from genotyping noise in the hard calls that a posterior threshold would collapse; the release carries no posteriors, so the blocks record the calls as released.

Two items of Sec. 3.5 closed with the build and two stay open. The parent assemblies are an ingestion dependency: each parent is pinned to its member of the 1011 assemblies tarball, and a parent absent from the member index raises. The cardinality cost was measured on the whole panel rather than on one Albert 2018 segregant. The intergenic question is not resolved but is no longer coupled to this class, since a mosaic is not a `GenePerturbation` and needs no gene; a cis-eQTL still has no gene to attach to, so per-variant attribution and the regulatory-DNA class remain one open decision. Ingesting the two matrices for Albert 2018 and Boocock 2025 is the natural next increment on the same genotype class.

References

[1] Brem, R. B., Yvert, G., Clinton, R. & Kruglyak, L. Genetic dissection of transcriptional regulation in budding yeast. *Science* **296**, 752–755 (2002). http://dx.doi.org/10.1126/science.1069516.

[2] Bloom, J. S., Ehrenreich, I. M., Loo, W. T., Lite, T.-L. V. & Kruglyak, L. Finding the sources of missing heritability in a yeast cross. *Nature* **494**, 234–237 (2013). http://dx.doi.org/10.1038/nature11867.

[3] Boocock, J. *et al.* Single-cell eQTL mapping in yeast reveals a tradeoff between growth and reproduction. *eLife* **13** (2025). http://dx.doi.org/10.7554/eLife.95566.

[4] Albert, F. W., Bloom, J. S., Siegel, J., Day, L. & Kruglyak, L. Genetics of trans-regulatory variation in gene expression. *eLife* **7** (2018). http://dx.doi.org/10.7554/eLife.35471.

[5] N'Guessan, A., Tong, W. Y., Heydari, H. & Nguyen Ba, A. N. Refining the resolution of the yeast genotype–phenotype map using single-cell RNA-sequencing. *eLife* **13** (2025). http://dx.doi.org/10.7554/eLife.93906.

[6] Smith, E. N. & Kruglyak, L. Gene–environment interaction in yeast gene expression. *PLoS Biology* **6**, e83 (2008). http://dx.doi.org/10.1371/journal.pbio.0060083.

[7] Nguyen Ba, A. N. *et al.* Barcoded bulk QTL mapping reveals highly polygenic and epistatic architecture of complex traits in yeast. *eLife* **11** (2022). http://dx.doi.org/10.7554/eLife.73983.

[8] Mancera, E., Bourgon, R., Brozzi, A., Huber, W. & Steinmetz, L. M. High-resolution mapping of meiotic crossovers and non-crossovers in yeast. *Nature* **454**, 479–485 (2008). http://dx.doi.org/10.1038/nature07135.